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An AI-Based Quiz System For Personalized Learning

Anonymous submission

Abstract

The AI-based quiz system (*iQS*) was designed to be used inside the most popular learning management systems (LMS) such as Moodle or MOOCs. The quiz system consists of four core components, viz. quiz content creator, quiz generator, knowledge graph manager and LMS quiz plug-in. The algorithms of these components can be reused directly, except that specific domain knowledge and specific learners' learning data need to be replaced in order to run in a new specific application domain. As a research instance of such a quiz system, a component for AAAA was developed and tested inside the Moodle learning platform of a university.

1 Introduction

The impact of Artificial Intelligence (AI) on learning, teaching and education is significant, unquestioned and ongoing (Chaudhry and Kazim 2022; Tuomi 2018; Seo et al. 2021), especially in supporting online learning and teaching, including personalizing learning for students, providing feedback to students' learning outcomes, contextualized learning for students, revolutionizing assessments, reducing teachers' workload, and developments in intelligent tutoring systems (ITS). With the same view as (Tuomi 2018), our proposed AI-based assessment system, which mainly has the form of quizzes, fully considers all three types of AI (i.e. logic-based, knowledge-based, and data-based neural AI), and focuses on the assessment of students' knowledge competence levels rather than accumulated semester credits.

For the currently digitalized online learning, personalized learning (Walkington and Bernacki 2020; Tang and Wang 2018) and self-regulated learning (SRL) are highly discussed approaches. Moreover, it is said that the most effective learners are self-regulating (Butler and Winne 1995). In general (Panadero 2017), a self-regulated learner normally monitors, directs, and regulates actions towards goals of information acquisition, expanding expertise, and self-improvement. Therefore we argue that digital learning support context should be goal-oriented and facilitate motivation and self-reflection. Feedback can have a powerful influence on SRL processes and affects students' cognitive engagement with tasks (Lim et al. 2021; Wisniewski, Zierer, and Hattie 2020). This study is also committed to study individual learning differences of students, differences of their learning stages, and how to provide the most adaptive and

accurate feedback at real time to improve students' learning and SRL skills.

Our research focuses on AI technologies applied in higher education and their practical applications, and further, on which innovations are needed to address emerging challenges in real scenarios. The target group for the development of the AI quiz were the students of a university, who are enrolled in distance learning courses using blended learning. Over half of them study part-time, having jobs or families, learning for a better career or certain interests, and are mainly learning by themselves. An AI-based online learning system could prove a powerful tool to support and facilitate their self-regulated learning processes (Lorenzo, Antonio, and Laura 2012; Luckin 2017).

In the initial phase of our project, we found that the learning management system (LMS) currently in use only records learning progress digitally in a coarse-grained way, allows to digitize learning content merely on a small scale and personalised input for students is limited to direct exchanges with tutors. Therefore, the data sets collected for specified data analysis are generally incomplete and fragmented or even missing. In addition, there are only a small number of tests that are manually created by tutors with fixed and general feedback and made available to all students without distinction.

After reviewing the current states of assessment/quiz systems, even some commercial ones (i.e. Questgen, Quillionz), we could not find one that is a fundamentally and systematically designed AI-based quiz system, also ready to use and easy to be extended with new AI capabilities. In order to meet our requirements for personalized learning and knowledge self-assessment, we built a scalable AI-powered quiz system.

Initially in the study, we stated the following research questions:

- Q1: How does a pervasive, scalable, and AI-based assessment/quiz system look like?
- Q2: How does it provide personalization and intelligent feedback at real time?
- Q3: How do students experience the tools support for self-regulated learning and does it motivate them to engage with the related learning material?

This study tries to answer the above-stated research ques-

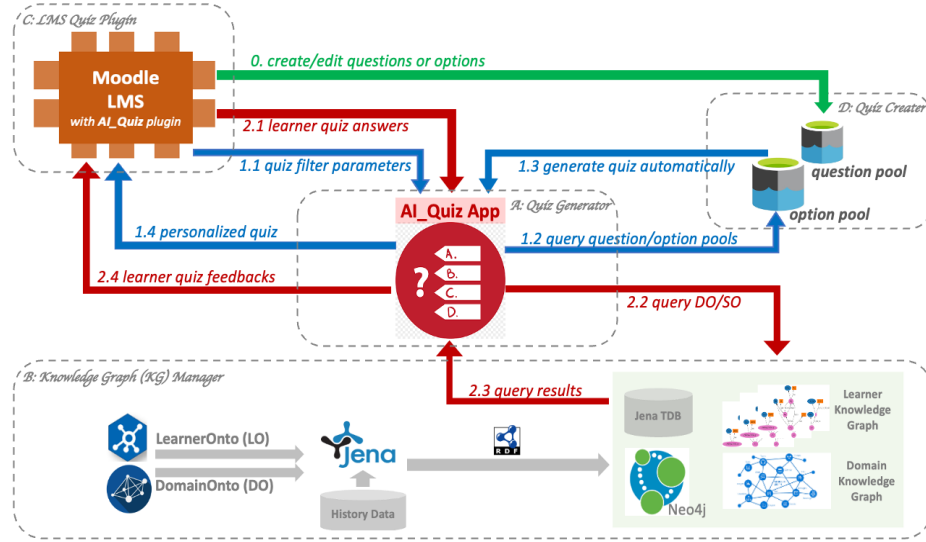


Figure 1: The AI-based quiz system (*iQS*)

tions, but is not limited to them. Such an assessment system, in the form of quizzes, is expected to be ready to use, intelligent, and easy to be scaled up with different machine learning (ML) models from different approaches (knowledge-based or learning analysis (LA) based) as well.

We have implemented a research instance of such a quiz system, named AAAA (?), according to the proposed theoretical framework (see Section 4). It has been preliminary tested and evaluated; the results obtained are presented in Section 6.

The first half of our research focused on two quiz generation algorithms and knowledge feedback. The corresponding results were published in (?), where our quiz system is described from the viewpoint of learning. However, this paper presents the whole AI-based quiz system and explains its innovation from the teaching point of view, which is, how a tutor creates a quiz, how much effort is involved, and which support he/she can get.

2 Related Work

Popenici and Kerr (2017) explored the phenomena of the emerging use of artificial intelligence in teaching and learning in higher education and deemed AI as part of the fabric of universities. The review of (Munir, Vogel, and Jacobson 2022) studied the use of AI-based algorithms in digital education, finding evidence that machine learning and deep learning algorithms are used in several themes of digital learning, such as intelligent tutors, performance predictions, adaptive and predictive learning and learning styles, analytic and group-based learning, automation and so on. Therefore, a full-fledged AI powered LMS system is a major interest and challenge in the area of higher education.

AI-based assessment systems have attracted attention and discussion and became very popular (Swiecki et al. 2022; Luckin 2017; Elshafey and Ghoniemy 2021). *AIAssess* is an

intelligent assessment software designed for students learning science and mathematics (Luckin 2017). We share a similar design philosophy in which we both consider, digitalising conceptual knowledge, offering students differentiated questions as tasks of increasing levels of difficulty as the student progresses, providing different levels of feedbacks as assistant, assessing each student’s knowledge of the subject matter, as well as their metacognitive awareness, knowledge of their own ability and learning needs, and having a knowledge component. However, we could not find any open source resource regarding *AIAssess* that could be used directly in our scenario. In addition, our work has innovated the structure of the quiz itself, so that our tests can be truly personalised and adaptable (see Section 3). This also has led us to consider making our own work and code publicly available as an open source project in the near future. Furthermore, an open-source interactive Quiz Maker and Management System (QMMS) has presented by (Elshafey and Ghoniemy 2021). They fully emphasized the importance of quiz to a LMS system and implemented a more effective and reliable system than Moodle from the software engineering viewpoint. However, they focus on quiz interactivity rather than how to create quizzes based on the personal level of knowledge.

Some AI-powered quiz platform on the market, such as Quillionz¹ and Questgen², focus on generating questions from the entered content or text with user manually specifying keywords or editing them afterwards. These platforms can server a certain level of assessments and quizzes for some casual learning, however, they fall far short of the requirements of generating quizzes from formal course content at universities or schools, let alone individualised quizzes based on students whole learning history.

¹<https://www.quillionz.com/>

²<https://www.questgen.ai/>

There are many research on generating quizzes and answer candidates and some are multiple choice question (MCQ) related (Ni et al. 2022; Qiu, Wu, and Fan 2020). Vachev et al. (2021) proposed a model that can generate a specified number of answer candidates for a given passage of text, which can then be used by instructors to write questions manually or can be passed as an input to automatic answer aware question generators. Liang et al. (2018) investigate how machine learning models, such as feature-based ensemble learning methods and NN-based, specifically ranking models, can be used to select useful distractors for multiple choice questions.

Although mainly focusing on rating the quality of answers to multiple-choice questions (MCQ), (Ni et al. 2022) pointed out a quite similar challenge to the one we are facing, namely, the deep implicit semantic relationships among learners, resources, courses and quiz questions. Unlike their work, we focus on how to construct real-time quiz questions to serve personalized and self-regulated learning of individual learners.

3 Two Semantic Premises

Our quiz system is built on top of two basic significant mechanisms, namely, a Domain Knowledge (DK) and an innovative Quiz Ontology (QO). Obviously, well defined domain knowledge with rich semantics on common prior or implicit intuitions is the key to AI applied in many application domains (Agarwal, Mishra, and Kolekar 2022; Zhuang et al. 2017).

In the project, we had more than 3 tutors as domain experts who went through at least 4 courses modules and their auxiliary textbooks, and manually built 4 course ontologies. They have not only defined the key knowledge concepts with very fine granularities, but also linked them to concrete learning material in various learning formats (i.e. text, audios, videos, slides, images). Such a domain knowledge lays the foundation for our work, enabling us to track and precisely locate students' knowledge errors, and to calculate their knowledge mastery distribution (i.e. what knowledge a student has learnt and how well he/she masters it, see Fig. 3). Therefore, we are able to provide intelligent knowledge-based recommendations to a specific student's current learning.

The second cornerstone is our Quiz Ontology. It comprises three innovations: (1) a quiz question is defined as one question (a.k.a, question body) and a set of options. The separation of question body and its options makes it possible to dynamically form personalized questions in an extremely flexible way. One option can loosely or semantically link to multiple questions, and vice versa. Questions and options are saved in different data pools, respectively. (2) A quiz will remain abstract until a set of concrete questions is to be formed on the fly. (3) Each option is required to semantically link to one or multiple knowledge concepts and learning materials, i.e. only when all its options are specified, a question will be clear about what concrete knowledge it is testing. This design can support quite many types of questions such as *Multiple Choice*, *Single Choice*, *True/False*, *Matching/Ordering*, *Image Drag/Drop*, or *Gap Select*.

In summary, designing quiz questions this way can guarantee the following: (1) different students get different quizzes, (2) the same student gets the same question body but with different options at different times, and (3) any quiz is generated fully based on students' own knowledge mastery levels at that very moment (for more see (?)).

4 The AI-based Quiz System

Turning to the quiz system itself, first, we briefly present its overview and the relationship among its four components, and then we address each of them in detail. The overall architecture of *iQS* consists of four major components:

Component A: the **quiz generator (QG)** dynamically and adaptively generates individual quizzes on the fly based on a set of real-time personal quiz filters, and provides knowledge-based feedback or recommendations instantly to students' answers (see Section 4.1).

Component B: the **knowledge graph manager (KGM)** manages the whole domain knowledge (DM), generating and updating learners' personal knowledge graphs (pKG) when tracking students' learning processes and results (see Section 4.2).

Component C: the **LMS quiz plug-in (QP)** enables *iQS* to be seamlessly embedded into concrete learning management systems (see Section 4.3).

Component D: the innovative **quiz creator (QC)** is responsible for semi-automatically creating quiz question building blocks, i.e. quiz question bodies and question options (see Section 4.4).

Fig. 1 illustrates the working process (following the blue lines) from the students' point of view: (1.1) students can take the system-suggested filter parameters as default, or they are free to change or set their own parameters to meet their quiz purposes (e.g. targeting on specific knowledge areas or the topics where they did wrong before, see the quiz filter in Fig. 2); (1.2) taking students' filters as input, sending queries to the question pool and option pools, and applying the defined quiz question generation algorithm (?) concrete quiz questions are formed for individuals; (1.3) individual quiz questions are generated; (1.4) the personalized quizzes are presented to students inside the quiz plug-in.

The underlying core idea of *iQS* is, at this moment, to understand who exactly learns what or does not, and at what competence mastery level. Therefore, the more *iQS* is used, the more accurate and intelligent the default filter becomes, since *iQS* will constantly track and analyze students' learning and learning results. Afterwards, it can go further to understand more, such as when and who does or does not do which activities in order to achieve certain mastery levels.

As mentioned above, the first three of the components considered in the sequel were developed previously (?). We summarize them briefly here to ensure the completeness of this paper, and also try to view them from a different angle.

4.1 The Quiz Generator (QG)

The innovation of the quiz generator component lies in its two core algorithms, one for generating personal quizzes

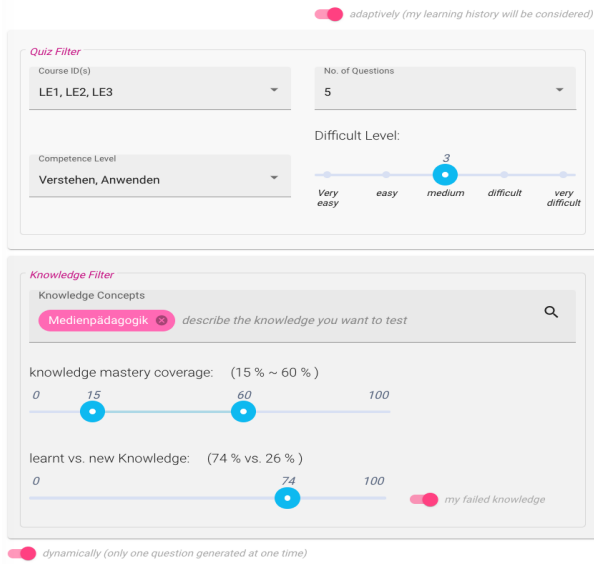


Figure 2: The personal quiz filter with 10 parameters

(named PQ), and the other one for providing intelligent knowledge-based recommendation for students' current learning situations or next step learning (named KR) (?). For example, to generate a personal quiz for a student i from a group of k students, the PQ algorithm requires the following inputs:

- a vector of 10 filter parameters being both quiz and knowledge related,
- a matrix of options, with n options in the option pool and each option having m linked knowledge concepts,
- a vector of knowledge concepts, with m knowledge concepts and each having a difficulty level value,
- the student's i learning cube over quizzes (SQ), where the entry SQ_{ij} tracks his/her latest three learning records for quiz option j ($j \leq n$); and each learning record capturing the information whether options were answered successfully and the time required for answering,
- a vector of the student's knowledge masteries, tracking the student's mastery levels of each knowledge concept.

Basically, the quiz filter takes in the parameters, and after a series of data normalizations, transformations and a matrix multiplication (defined in (?)) the final winner options will stand out. Due of the semantic links between question bodies and options, the candidate questions are substantially assembled as well eventually.

On the other hand, the KR algorithm is fully based on a set of rules (defined in (?)) to generate an instant knowledge feedback. It distinguishes between the two cases whether an answer received is correct or incorrect. When a student submits an incorrect answer, the feedback will focus on how to correct and improve his/her current knowledge level. Conversely, when his/her answer is correct, then the system will recommend new further knowledge for him/her to learn from there. Moreover, since each option semantically links

to multiple knowledge concepts, by cross-checking the student's answer options, it is possible for us to accurately detect the exact position of the student's knowledge errors, if any.

More importantly, tests with two simulated datasets showed that the algorithms are quite efficient, as it takes less than 2 seconds to generate a quiz with 30 questions for a student on both the small dataset and the big one (see more in (?)). The simulated big dataset contains some 300 key knowledge concepts, 10K students, and 100K options; the ratio of one option to belong to multiple questions is temporarily kept around 10~15%.

4.2 The Knowledge Graph Manager (KGM)

Knowledge tracking is crucial and is about inferring the skill mastery of students and predicting their performance to adjust the learning accordingly. Since knowledge concepts and difficulty levels have been specified inside questions, individual knowledge mastery are able to be tracked in our system, which conforms to the requirements of the interpretable knowledge tracking model proposed by (Minn et al. 2021). Therefore, the tracking learning records collected by our system can be used for further deep learning analytic.

The KGM component is supposed to not only manage the domain knowledge, but also to be responsible for creating and maintaining the students' individual knowledge graphs. For example, after a while of self-regulated learning, the personal knowledge graph of student s will be updated as shown in the Fig. 3. The green nodes represent the knowledge that he/she has learnt, and the gray ones represent what still has to be learnt. The size of the nodes is used to distinguish mandatory from optional knowledge. Each node is also associated with a knowledge mastery value; for example, 0.45 means that the mastery level 45% is currently achieved on this knowledge point. In addition, a small summary report is attached to show the student's personal progress. For example, compared with his/her previous quiz history, this student's knowledge mastery rate has increased by +3.0% after the current quiz.

4.3 The Quiz Plug-in (QP)

The quiz system iQS is designed for general purposes and versatile domains from the very beginning. Particularly for the current environment of micro-services and containers as mainstream application architectures, we fully take into account scalability, flexibility, and loose coupling of our micro-services. Therefore, our components can be wrapped up as plug-in micro-service for any current popular LMS, such as Moodle³ or MOOCs⁴.

Since our current testbed is a Moodle learning system, a research instance of iQS was implemented as a Moodle plug-in. Preliminary assessment results are presented in Section 6).

³https://docs.moodle.org/dev/Moodle_App_Plugins_Development_Guide

⁴<https://www.mooc-list.com/tags/plugin-development>



Figure 3: Example of a learner's personal knowledge graph (pKG)

4.4 The Quiz Creator (QC)

As the focus of this paper, the Quiz Creator component represents our innovations in terms of teaching. It is completely different from traditional quiz creation, where all students take exactly the same quiz. Such a quiz cannot be adaptive to students' learning states nor can it dynamically interact in real time. Our AI-based *QC* aims not only to free tutors from heavy manual work, but also from impossible tasks (e.g. big mounts of real-time personal quizzes with intelligent feedback), allowing them to pay more attention to the design of teaching content itself.

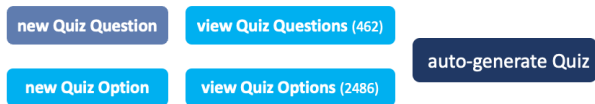


Figure 4: Five modules of the Quiz Creation component

The *QC* contains five modules (see Fig. 4). The modules *new quiz question* and *new quiz option* are used to create questions and options, respectively; the modules *view quiz question* and *view quiz option* allow to view existing questions or options. The last module *auto-generate quiz* is used to automatically generate quiz options from a domain knowledge and learning resources, for example, from text books.

5 Evolutionary Process of Quiz Creation

To get *iQS* started, certain collected quiz data need to be entered manually. Later, these data can also be used as golden standard for data analytics. Therefore, the *QC* component will go through two development phases, one involving manual work and a fully automatic one.

5.1 Instructor-involved Quiz Question Creation

Different from existing quiz systems, multiple tutors can operate *iQS* at the same time, and the created questions and options are directly saved to the question and option data pools, respectively.

Only very few inputs are required to create a question. For instance, in Fig. 5, a tutor *T* only needs to set the type of question, the score, the title and the question body. The remaining fields will mostly be auto-filled once some options have been linked to the specific question. For example, the contents of fields *course ID(s)*, *knowledge concepts*, *competence level* and *difficulty level* will be sums or maximum values of the corresponding information of all linked options.

There is also the possibility to specify any question option at this point. The three ways to link an option to the current question are: (1) to search for one in the option pool; (2) to create a completely new one; (3) to browse the system-suggested ones by clicking on the number next to the button (*auto-suggested*) *correct options*. Last but not least, a tutor also can provide a short text as content of *further recommendation* in case the question was answered correctly. Such short texts are collected and later used to fine-tune our NLP-based feedback generation algorithm.

Similarly, there are very few inputs required when creating an option. For instance, in Fig. 6, a tutor can independently create a question option by defining the option body, the knowledge concepts to be tested, the difficulty level, the competence level, and short texts as feedback to correct and/or incorrect answers. At this stage the tutor also has the opportunity to manually link the option to existing questions using the search function. Notably, when an option is assigned to a specific question body, two pieces of information have to be given, that is the correct answer to this question and the score weight.

Figure 5: Example of creating a new question and managing its options

5.2 Automatic Quiz Creation

Automatically creating questions and options to fill in the two data pools from raw course material will be our next step. Since automatically generating quizzes is already a well researched topic, quite some work was done in this area by fusing technologies of neural networks (NNs) and natural language processing (NLP) (Gupta et al. 2021; Kurdi et al. 2020; Liang et al. 2018; Nwafor and Onyenwe 2021; Qiu, Wu, and Fan 2020). For example, (Gupta et al. 2021) uses NLP and optical character recognition (OCR) to extract keywords from uploaded text images (e.g. scanned books) and from the internet to generate facts-based multiple choice questions (MCQ).

Moreover, (Liang et al. 2018; Nwafor and Onyenwe 2021; Qiu, Wu, and Fan 2020) worked on generating distractors of MCQ from free text with different methods, such as a point-wise ranking support vector machine, a list-wise ranking neural network, and a ranking generative adversarial network (GAN). The results of their experiments are very inspiring as well. We may consider to directly apply similar approaches in our context in the next step.

6 User Experience and Motivation

As a research instance of *iQS*, AAAA is still under development and has just been put into use. Hence, we are not

Figure 6: Example of creating a new option and manually linking to a question

yet able to evaluate its capabilities in depth, such as whether the quizzes generated fit the individual situations and needs. However, initial user experiences (UX) are already being gathered. Although the following evaluation results cannot reflect the entire potential and effectiveness of our work, they will give insight into what might happen when AI applications enter into actual educational settings.

To evaluate different stages of the AAAA prototype throughout the development process, an iterative cycle was chosen including tutors and students from an educational science course as well as subject matter experts. This enabled monitoring of the quality of each developmental stage and feedback provided by tutors, students and experts could inform further improvements. Initially, the AAAA-prototype v0.1 was evaluated by 7 tutors using two surveys. The first survey was designed in-house and assessed the general experience with the feedback and technical information. The feedback was predominantly regarded as relevant and helpful. Suggestions for further development from open-ended questions focussed on an greater enrichment of the feedback. The second survey was the UX survey Attrakdiff (Hassenzahl, Burmester, and Koller 2008). The AAAA was considered to have a medium to high perceived hedonic and pragmatic quality. The classification fluctuated between raters in the areas desired, action-oriented, neutral and self-oriented, showing a tendency for positive UX ratings.

The analysis of this initial evaluation was used to further develop the AAAA prototype v0.2. It was presented to students in the winter semester 2021/2022 via weblink inside a Moodle research instance. 512 students had access to the AAAA for a duration of 17 days. In that timeframe 111 quizzes were created by 69 students. The widely used and reliable system usability scale (SUS) (Brooke 1996) was

used to assess the AAAA. The SUS contains 10 items accumulated for a final score. Standardized interpretations of the scores consist of percentage based ratings or grades from A+ to F (Lewis and Sauro 2018). In addition to the SUS, the student evaluation included two items to assess the satisfaction with the AAAA and students' perceived effect on motivation to engage related learning material. A 5-point Likert scale (from 0 – strongly disagree/very unsatisfied to 4 – strongly agree/very satisfied) was used to rate the items. Whenever participants chose poor ratings (0,1), optional free text questions were presented, to enable an elaboration of the rating.

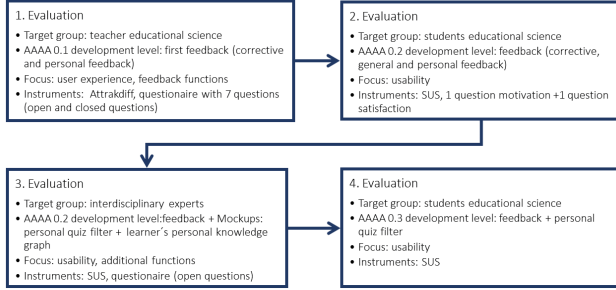


Figure 7: AAAA evaluation process

The AAAA was rated by 35 participants with high usability ($M = 76.28, sd = 15.64$), corresponding to a grade of B. Additionally they assessed the AAAA as motivating, to engage with the learning material ($M = 2.94, sd = 0.80$) and were satisfied with the current state of the AAAA ($M = 3.09, sd = 0.51$). To extend the second iteration of the evaluation in January 2022 the SUS was presented to experts from the fields of educational science, psychology, social sciences and information technology. For this second iteration the AAAA-prototype 0.2 was extended with mockups for planned developments, including additional visual feedback in the form of a learner's personal knowledge graph and a personal quiz filter page to define parameters that students can use to influence the quiz generation. The expected drop in rating due to these not yet completed components of the ISFE showed up in the usability rating. The 11 participating experts rated the AAAA with low usability ($M = 56.59, sd = 20.90$), resulting in a grade of D. Scores below C (68 points or lower) mostly referred to problems understanding the presented concepts and constructs in the mock-ups, explaining why the experts were more critical of the usability of the AAAA than students (Cohen's $d = -1.07$). The analysis of the results led to decreasing complexity and implementing mouseover information for the different functions.

	AAAA v0.2 ($n = 35$)		AAAA v0.3 ($n = 46$)	
	M	sd	M	sd
Students	76.28	15.64	78.10	12.07
Experts	56.59	20.90		

Table 1: SUS score by AAAA versions

The AAAA-prototype v0.3 was then implemented as a

Moodle plug-in inside the research instance. Smaller improvements to the visual design were made during this phase. The fourth evaluation took place in the summer semester 2022. 771 students had access to the AAAA plug-in for the duration of 39 days. In that timeframe 119 quizzes were created by 46 students. 46 students answered the questionnaire, again resulting in a high usability rating for the plug-in ($M = 78.10, sd = 12.07$) equivalent to the grade B+. As in the previous assessment, students felt the AAAA was motivating them to engage with the learning material ($M = 3.13, sd = 0.78$) and were satisfied with the tool ($M = 2.83, sd = 0.77$).

	AAAA v0.2 ($n = 35$)		AAAA v0.3 ($n = 46$)	
	M	sd	M	sd
Motivation	2.94	0.80	3.13	0.78
Satisfaction	3.09	0.51	2.83	0.77

Table 2: Students ratings on motivation and satisfaction.

While the results of the last evaluation are promising, further developments are currently being integrated and could not yet be evaluated. A teacher page was developed and implemented, enabling teachers to add to the question pool, whilst the system automatically integrates their input into the ontology. Additionally, an in depth evaluation of the effects of using the AAAA in experimental settings with added standardized psychometric instruments to measure motivation (e.g. SMR-LS, see Thomas and Müller 2016) and satisfaction (e.g. PYTHEIA, see Koumpouros, Papageorgiou, and Karavasili 2017) is being prepared.

So far, three versions of AAAA-prototypes (v0.1~0.3) were evaluated based on their initial usage. Although we were prepared for some criticism due to incompleteness of functionality in the initial versions, the results are rather positive. To reflect specifically on the more critical feedback we received, influencing factors are possibly: (1) the system at this stage focuses on the provision of functionality rather than high usability; (2) additional information on the systems usage (e.g. more communication between users and system designer, user documentation, tutorials) has not yet been introduced. There is room for improvement on these aspects.

7 Conclusion

A novel AI-based quiz system was presented in detail, which aims to support students' self-regulated learning with personalized quizzes. It is supposed to be a pervasive, scalable, and AI-based assessment system to be used in various application domains. As first implementation, the AAAA system was tested and evaluated by university students and teachers rendering a first batch of very objective feedback. Students showed certain satisfaction and stated that AAAA could motivate them to engage more with the learning material than before. Compared with students, experts' acceptance is relatively low at this moment.

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